Product Requirements Document: Image Analyzer

Image Analyzer

Status: Describes the current project implementation **Product type:** Desktop application **Primary entry point:** `image_caption_generator.py`

1. Product summary

Image Analyzer is a desktop tool for exploring and describing a single image at a time. It combines local digital image processing with AI-powered captioning and image question answering. Users can choose a cloud model for richer features or a locally run BLIP model for basic captions.

The product is intended as an accessible Digital Image Processing project and a practical utility for quickly understanding, describing, and exporting information about images.

2. Problem statement

People often need a quick description of an image, a basic summary of its visual properties, or an answer to a question about its contents. Separate tools are typically needed for image statistics, captions, and visual questions. This application puts those tasks behind one desktop interface and supports both cloud-based and local caption generation.

3. Goals

- Let a user load or paste an image and see it in a desktop application.
- Provide short and detailed AI-generated descriptions.
- Provide local captioning when the BLIP model is installed and available.
- Show understandable image statistics and visualizations.
- Allow users to ask questions about an image and request a prediction-style caption.
- Make results easy to copy, save with the image, and hear aloud on Windows.
- Present useful errors for missing credentials, network failures, provider limits, and model availability.

4. Non-goals

The current product does not provide image editing, batch processing in the GUI, image search, persistent project/history storage, model training, or guarantees that AI interpretations are correct. The command line does support captioning multiple paths, but there is no batch management interface.

5. Target users

- Students learning digital image processing and computer vision concepts.
- Users who want quick image descriptions or simple image statistics.
- Users who need a basic local captioning option after the BLIP model has been downloaded.

6. User workflows

6.1 Load and inspect an image

1. Start the application.

2. Open an image with the upload control, click the preview, or paste an image with `Ctrl+V`.
3. View the image preview and image-analysis tab.
4. Review dimensions, brightness, contrast, dominant colors, and the RGB histogram.

6.2 Generate a caption

1. Load an image.
2. Select the online or offline model and choose short or detailed output.
3. Generate or retry the caption.
4. Copy the caption, read it aloud where supported, or save a captioned image.

6.3 Ask a question or generate a prediction

1. Load an image.
2. In the question tab, enter a question or choose an example and submit it; or open the prediction tab and request a prediction caption.
3. Review the result and copy or read it aloud. Prediction captions can also be saved with the image.

7. Functional requirements

ID	Requirement	Current implementation
FR-1	Accept an image from a file picker and clipboard paste.	Supports JPG/JPEG, PNG, BMP, GIF, WebP, TIF/TIFF; `Ctrl+O` and `Ctrl+V` shortcuts.
FR-2	Normalize image orientation and color for processing.	Applies EXIF orientation and converts to RGB, compositing transparency on white.
FR-3	Display an image preview and analyze the loaded image.	Shows dimensions, brightness, contrast, five dominant colors, and an RGB histogram.
FR-4	Generate short and detailed captions online.	Sends a resized image to Groq, with Gemini fallback.
FR-5	Generate captions locally when optional dependencies are installed.	Uses Hugging Face BLIP; downloads model weights on first use if necessary.
FR-6	Answer user questions about the image.	Online vision request, with answers intended to be one or two short sentences.
FR-7	Generate a prediction-style caption.	Online vision request that makes cautious inferences from visual clues.
FR-8	Export and share generated text.	Copy controls for caption, answer, and prediction; save caption and prediction below the image.
FR-9	Read generated text aloud.	Windows System.Speech for captions, answers, and predictions.
FR-10	Communicate errors and provider status.	Converts common network, credential, model, and rate-limit failures into user-facing messages; retries brief rate limits once.
FR-11	Support command-line captioning.	Accepts image paths, `--offline`, and `--predict` options.

8. Non-functional requirements

- **Usability:** Core actions should be available from a desktop GUI with keyboard shortcuts for opening and pasting.
- **Responsiveness:** Network and model work should not prevent the interface from handling normal input while processing.
- **Privacy awareness:** Online features transmit an image to a third-party provider. Users should be able to choose offline captioning for the supported caption workflow.
- **Resilience:** Provider errors and rate limits should be surfaced clearly, and the online path should try a second configured provider when possible.

- **Compatibility:** The application is built with Python, Tkinter, Pillow, and Requests. Windows is the supported experience for built-in speech; other platforms may run the GUI and image processing if Tkinter and dependencies are available.
- **Maintainability:** Provider model IDs and image size limits are centralized near the top of the application module.

9. Data, integrations, and dependencies

- **Image formats:** JPG/JPEG, PNG, BMP, GIF, WebP, TIF/TIFF (subject to Pillow support and the file contents).
- **Online providers:** Groq OpenAI-compatible chat completions and Google Gemini generateContent API.
- **Offline model:** Salesforce BLIP image-captioning base model via Hugging Face Transformers.
- **Image processing:** Pillow.
- **Desktop UI:** Python Tkinter/ttk.
- **Configuration:** `GROQ_API_KEY` and `GEMINI_API_KEY` may be loaded from a project-local `.env` file or process environment.

Images sent online are converted to JPEG and reduced so their longest side is at most 768 pixels. The local captioning path processes the image with BLIP and does not need an online provider key once the model files are available.

10. Acceptance criteria

- A user can open a supported image or paste an image and see it in the application.
- Loading an image updates the preview and its image-analysis values and plots.
- Online captioning uses an available configured provider and provides a clear message if no provider can answer.
- With the optional packages and BLIP weights available, offline captioning can produce a caption without an online API key.
- The question and prediction workflows return results through the online provider path.
- Copy and image-save controls produce the corresponding text or captioned image.
- Windows users can start and stop speech playback for generated text.
- The command line can caption one or more image paths and supports offline and prediction modes.

11. Constraints and known limitations

- Groq, Gemini, and Hugging Face availability, quotas, pricing, and model behavior are controlled by external services and may change.
- Online image questions and prediction captions require internet access and at least one valid API key.
- The first BLIP setup may require a large model download and sufficient local memory and storage.
- Image analysis uses simple pixel statistics; brightness and contrast are descriptive estimates, not quality judgments.
- Predictions about thoughts, feelings, intent, or future events are uncertain model inferences.
- The project is currently contained in a single Python module and has no committed dependency lockfile or automated test suite.

12. Future considerations

These are possible follow-on improvements, not claims about current behavior:

- Add automated tests for image normalization, analysis, provider error handling, and output saving.
- Add a dependency manifest and clearer first-run setup checks.
- Add configuration controls for model selection and online privacy.
- Add batch processing and a history/export workflow to the GUI.
- Improve accessibility, keyboard navigation, and cross-platform speech support.